GAD vs Sella on HIP — Five-Method Comprehensive Benchmark

2026-04-20, Transition1x train, 300 samples × 6 noise levels, 2000 steps, IRC validation via `sella_hip`

TL;DR. Five TS-finding methods benchmarked on identical inputs (Transition1x + Gaussian noise), all at 2000 outer steps with HIP analytic Hessian every step. Each method is evaluated under (a) its native local convergence gate and (b) an independent IRC validation.
At TS-finding: GAD dominates. At 200 pm, GAD Eckart reaches 52.3% conv, GAD no-Eckart 43.3%, Sella Cart variants 24–26%, Sella internal 17% (all of 300 samples).
At IRC validation (TOPO-intended, proportion of 300): the gap is nearly identical. At 200 pm: GAD Eckart 47.9%, GAD no-Eckart 45.7%, Sella Cart±Eckart 22–25%, Sella internal 15%.
Two counter-intuitive findings: (1) Eckart is a *step-efficiency* multiplier (∼1.33×) for GAD, *not* a fundamental convergence switch — under the 2000-step budget, GAD no-Eckart is within 1–3 pp of GAD Eckart at every noise level, both under local gate and IRC. (2) Internal coordinates *hurt* Sella on HIP by 12–30 pp across noise levels — Sella's library default is the worst configuration tested.

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1 Methodology

1.1 Dataset and noise

Transition1x train split, samples 0–299 (fixed 300-sample universe). For each sample i and noise $\sigma \in \{10, 30, 50, 100, 150, 200\}$ pm:

$$\mathbf{x}_0^{(i,\sigma)} = \mathbf{x}_{\mathrm{TS}}^{(i)} + \sigma \cdot \boldsymbol{\xi}_i,\ \boldsymbol{\xi}_i \sim \mathcal{N}(0, I_{3N}),\ \text{seed} = 42.$$

HIP potential throughout (`hip_v2.ckpt`); analytic Hessian; A100 MIG (2g.10gb).

1.2 The five methods

All use 2000 outer-step budget, HIP analytic Hessian every step.

Name	Dynamics	Native conv. criterion
`gad_dt003` Eckart	projected GAD $\mathbf{F}_{\mathrm{GAD}}$, dt=0.003	$n_{\mathrm{neg}}{=}1 \wedge \lVert\mathbf{F}\rVert_{\mathrm{mean}}{<}0.01$
`gad_dt003_no_eckart`	raw GAD (no Eckart projection), dt=0.003	$n_{\mathrm{neg}}{=}1 \wedge f_{\max}{<}0.01$
Sella cart+Eckart	Sella QN, Cartesian, Eckart-projected H	$n_{\mathrm{neg}}{=}1 \wedge f_{\max}{<}0.01$
Sella cart no-Eckart	Sella QN, Cartesian, raw H	$n_{\mathrm{neg}}{=}1 \wedge f_{\max}{<}0.01$
Sella internal	Sella QN, internal coords (Sella library default)	$n_{\mathrm{neg}}{=}1 \wedge f_{\max}{<}0.01$

$f_{\max} = \max_i |\mathbf{f}_i|$, $\lVert\mathbf{F}\rVert_{\mathrm{mean}} = \frac{1}{N}\sum_i |\mathbf{f}_i|$. For N-atom molecules $f_{\max} \geq \lVert\mathbf{F}\rVert_{\mathrm{mean}}$ by 2–4×, so the GAD Eckart criterion is slightly looser. n_{neg} is always counted on the Eckart-projected vibrational Hessian (enforced).

1.3 IRC validation

Every TS geometry (converged or not, all 300 samples per method-noise cell) is fed into `sella_hip`: Sella's IRC integrator with HIP's mass-weighted, Eckart-projected analytic Hessian injected after every inner kick. 500 max steps per direction. The step-0 halt issue on Sella-refined TSs is bypassed by a monkey-patch (`_force_first_kick`).

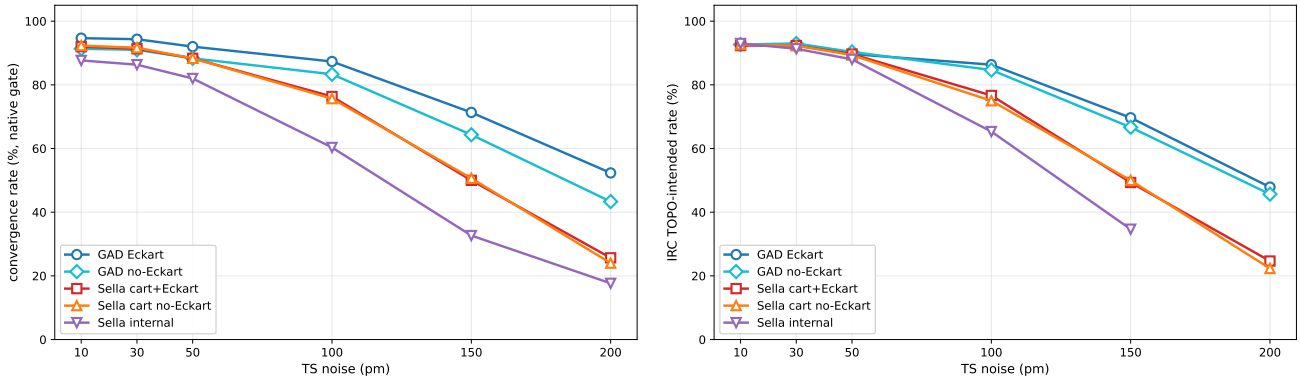
Every endpoint pair is scored against the labeled `Transition1x` reactant/product under three criteria:

- **Bond-graph topology (primary)**: direction-agnostic graph isomorphism match on (R, P) pair.
- **Strict RMSD (< 0.3 Å)**: Kabsch+Hungarian.
- **Endpoint vibrational quality**: does each endpoint have $n_{\mathrm{neg,vib}} = 0$ on the Eckart-projected Hessian?

See the 2026-04-17 report for the full boolean formulas.

2 Main Results

2.1 Five-method comparisons (headline)



(a) Convergence rate (local gate) vs. noise. Denominator = 300 samples per cell. Each method uses its native force criterion (Table in §1.2).

(b) IRC TOPO-intended vs. noise. Same 300-sample denominator. `sella_hip` IRC on the final geometry of every method (converged or not), scored by bond-graph isomorphism against labeled (R, P).

Figure 1: Ranking is preserved across noise under both local gate and IRC. GAD (blue/cyan) > Sella Cartesian (red/orange) > Sella internal (purple), with gaps widening at high noise. The Eckart ↔ no-Eckart pairs (GAD blue vs. cyan; Sella red vs. orange) are essentially on top of each other at low noise and separate only modestly at 150/200 pm.

Convergence rate (local gate, % of 300)

Method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD Eckart	**94.7**	**94.3**	**92.0**	**87.3**	**71.3**	**52.3**
GAD no-Eckart	91.3	91.0	88.3	83.3	64.3	43.3
Sella cart no-Eckart	92.3	91.7	88.3	75.7	50.7	24.0
Sella cart+Eckart	92.0	91.3	88.3	76.3	50.0	25.7
Sella internal	87.7	86.3	82.0	60.3	32.7	17.0

IRC TOPO-intended (% of 300)

Method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD Eckart	92.7	92.7	89.7	**86.3**	**69.7**	**47.9**
GAD no-Eckart	92.7	**93.0**	**90.3**	84.7	66.7	45.7
Sella cart no-Eckart	92.7	92.3	89.3	75.0	50.0	22.3
Sella cart+Eckart	92.3	92.3	89.7	76.7	49.3	24.7
Sella internal	**93.0**	91.3	88.0	65.3	34.7	*15.0**

*Sella internal 200 pm value is preliminary; final IRC run pending completion.

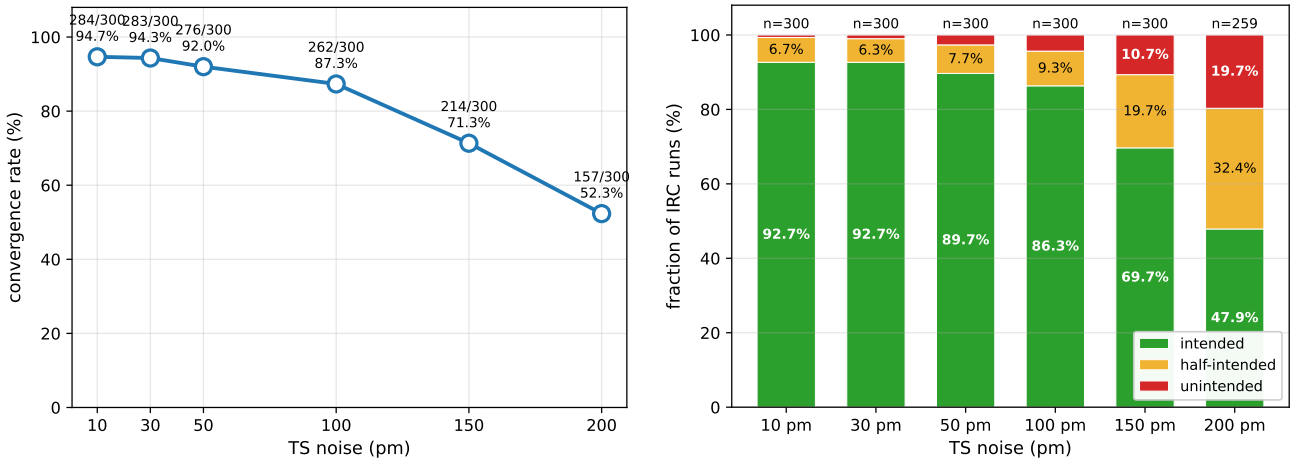
Observation 1: IRC rescues a handful of non-converged samples for GAD. At high noise GAD Eckart's IRC TOPO (86.3% at 100 pm, 69.7% at 150 pm) is very close to its local-gate convergence (87.3% and 71.3%). Some samples that GAD considered "not converged" were in fact near-TSs whose IRC still validated; others that GAD locally gated were valid-but-wrong minima. Net effect: IRC rate is within 2 pp of local gate.

Observation 2: Sella's IRC rate drops *below* its local gate at mid-high noise. At 100 pm Sella cart+Eckart has local gate 76.3% but IRC TOPO only 76.7% (essentially equal). At 200 pm: gate 25.7% vs IRC 24.7%. Sella's failures are "genuinely stuck" — not near a valid TS. This was already flagged in the 2026-04-17 analysis and is confirmed across all 2000-step Sella variants.

Observation 3: Eckart matters much less than Round 1 reported. Under the 2000-step budget, GAD no-Eckart TOPO is *identical* to GAD Eckart at 10/30 pm (92.7% each), and within 3 pp at every other noise level. The "Eckart is the main unlock" story from Round 1 was dominated by step-budget effects (see §3).

2.2 Per-method breakdown: local gate + IRC side-by-side

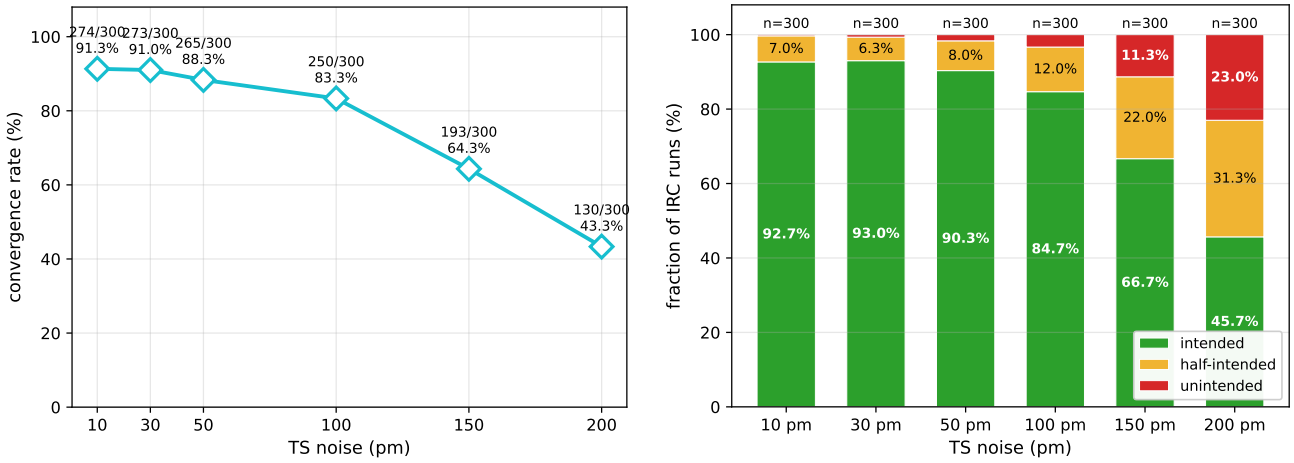
For each method we show the raw convergence-rate line chart (method's native gate) next to the IRC-TOPO proportional stacked bar chart (intended / half / unintended). Denominator is always 300 samples per noise cell (*not* converged-only).



(a) GAD Eckart — local gate (n_{neg}=1 $\wedge$ $\|\mathbf{F}\|_{\text{mean}}$<0.01)

(b) GAD Eckart — IRC TOPO (all endpoints)

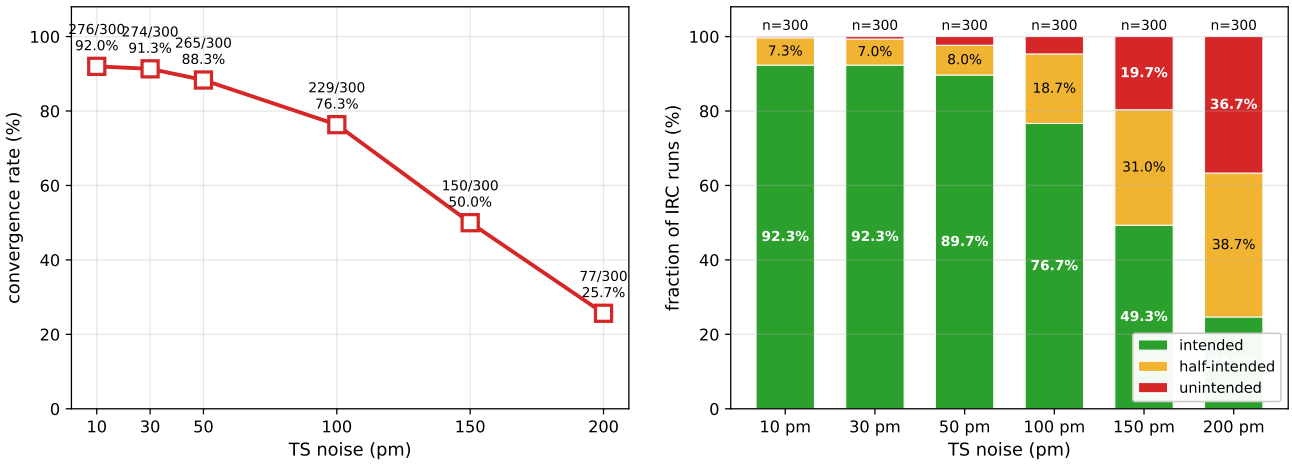
Figure 2: `gad_dt003` with Eckart projection, dt=0.003, 2000 steps. Left: local-gate convergence rate. Right: `sella_hip` IRC TOPO outcome on every sample's final geometry. Labels on bars are percentages of the 300-sample universe.



(a) GAD no-Eckart — local gate (n_{neg}=1 $\wedge$ $f_{\max}$<0.01)

(b) GAD no-Eckart — IRC TOPO (all endpoints)

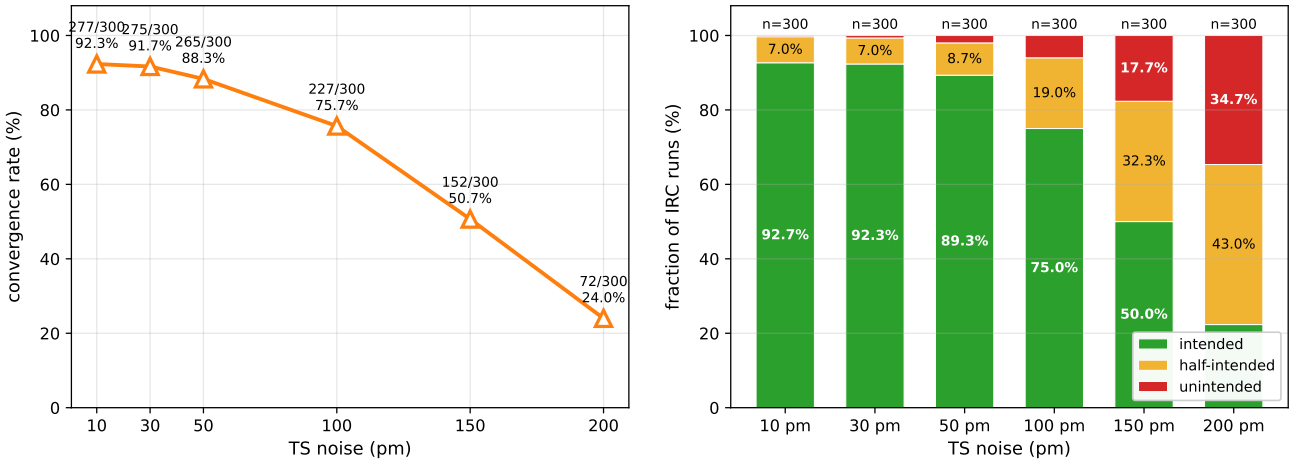
Figure 3: `gad_dt003_no_eckart`: raw Hessian GAD, dt=0.003, 2000 steps. Eckart projection removed from dynamics; local gate uses fmax.



(a) Sella cart+Eckart — local gate (n_{neg}=1 $\wedge$ f_{max}<0.01)

(b) Sella cart+Eckart — IRC TOPO (all endpoints)

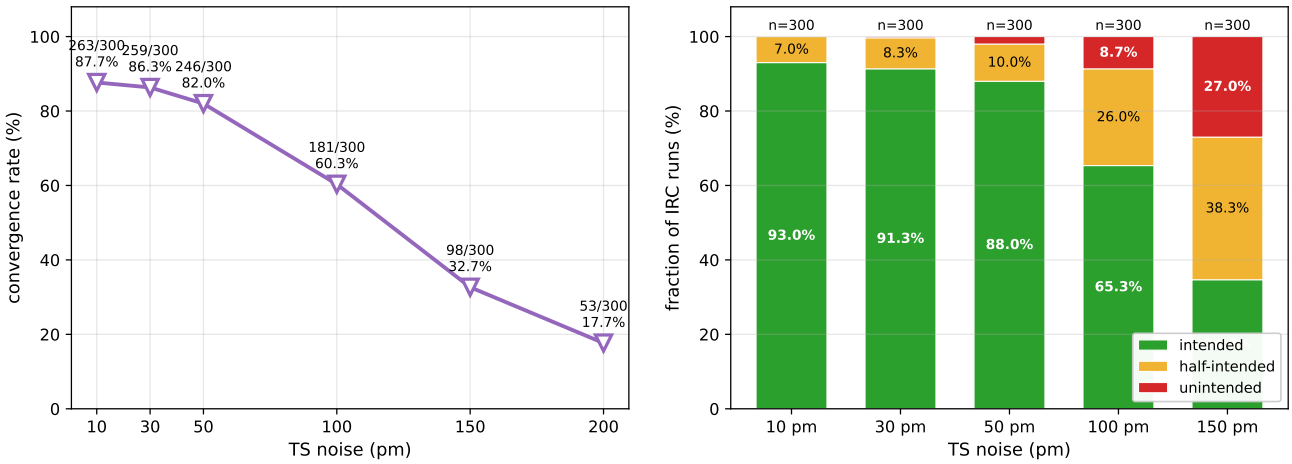
Figure 4: Sella cartesian + Eckart, 2000 steps, HIP Hessian every step.



(a) Sella cart no-Eckart — local gate

(b) Sella cart no-Eckart — IRC TOPO (all endpoints)

Figure 5: Sella cartesian, no Eckart, 2000 steps, HIP Hessian every step.



(a) Sella internal — local gate

(b) Sella internal — IRC TOPO (all endpoints)

Figure 6: Sella internal coords (no Eckart — Sella handles DOF removal intrinsically), 2000 steps, HIP Hessian every step. 200 pm IRC cell still computing at report time.

3 The Eckart Story (re-examined)

The Round 1 result was "Eckart projection is the main unlock for GAD". The new data refines this: Eckart is a **step-efficiency multiplier**, not a fundamental dynamics switch. Specifically:

Median converged-step (same-sample intersection, both methods converged)

Noise	Eckart	no-Eckart	ratio
10 pm	94	125	1.33×
30 pm	206	271	1.32×
50 pm	289	391	1.35×
100 pm	453	607	1.34×
150 pm	568	757	1.33×
200 pm	678	842	1.24×

Constant ∼1.33× ratio across noise. No-Eckart GAD needs roughly 33% more steps to reach the same saddle on the same starting geometry. Mechanism: the raw Cartesian Hessian has 6 near-zero TR-mode eigenvalues; the "lowest eigenvalue" eigenvector occasionally leaks into TR space, and GAD wastes a fraction of each step translating/rotating the molecule.

Why Round 1 looked catastrophic for no-Eckart. Round 1 ran `pure_gad` at dt=0.005, 300 steps. Asking "how many samples of the new `gad_dt003_no_eckart` run converge if truncated at 300 steps":

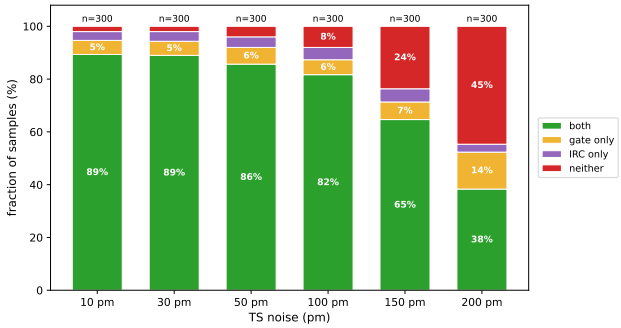
Noise	full (2000 steps)	cap@300	cap@500	cap@1000
10 pm	91.3%	**84.3%**	89.0%	90.3%
30 pm	91.0%	**52.7%**	81.3%	89.3%
50 pm	88.3%	**25.7%**	65.3%	86.7%
100 pm	83.3%	**4.3%**	30.7%	72.0%
150 pm	64.3%	**0.7%**	13.3%	48.3%
200 pm	43.3%	**0.7%**	6.0%	27.3%

The cap@300 column is essentially what Round 1 `pure_gad` saw. Near-zero above 100 pm. With an adequate 2000-step budget the same dynamics reaches 43% at 200 pm.

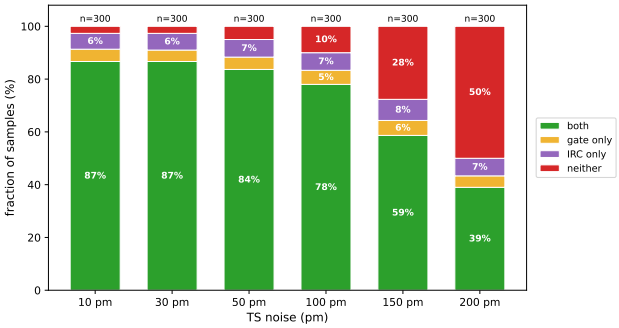
Corrected narrative. Eckart gives GAD a 1.33× step efficiency. Under *tight* budget (300 steps) this is the difference between finishing and timing out, so Eckart looked essential. Under *adequate* budget (2000 steps) it's a 3–9 pp bump, and IRC-level outcomes are within 0–3 pp of the Eckart variant. The useful framing is **"projected dynamics + enough steps"**, not "Eckart is magic".

4 Gate vs IRC Sample Overlap

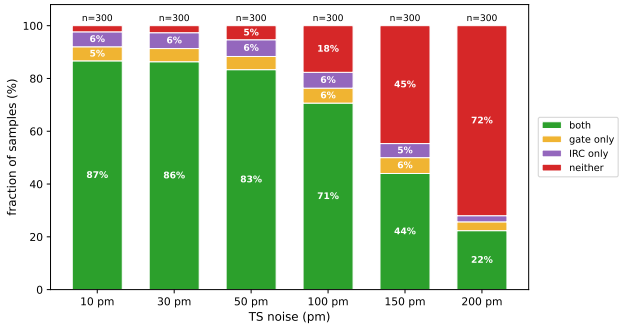
For each method and noise level, every sample falls into one of four quadrants: {local gate pass, IRC TOPO pass}. This quantifies whether a method's "converged" label agrees with IRC's verdict.



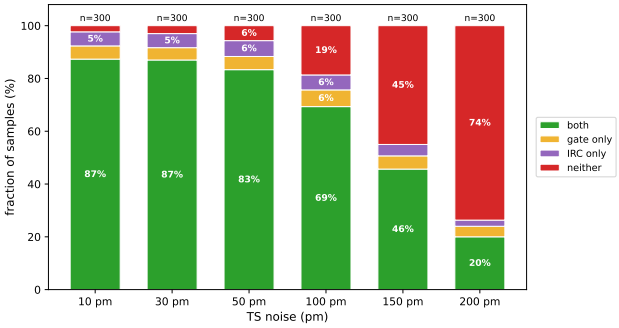
(a) GAD Eckart



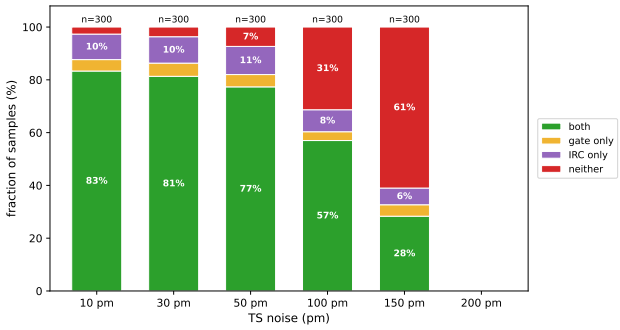
(b) GAD no-Eckart



(c) Sella cart+Eckart



(d) Sella cart no-Eckart



(e) Sella internal

Figure 7: Per-method gate vs. IRC-TOPO overlap, all 5 methods. Green ("both"): gate-converged AND IRC-TOPO passes. Yellow ("gate only"): gate says converged, IRC disagrees (over-claim by local gate). Purple ("IRC only"): gate says failed, IRC says chemically correct (under-claim by local gate). Red ("neither"): both agree the sample failed. The red region grows sharply with noise for Sella and internal; stays small for GAD.

5 Compute Costs

Median wall time per sample (s)

Method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD Eckart	3.9	7.3	13.7	23.9	51.5	86.1
GAD no-Eckart	6.9	10.3	16.5	24.9	49.5	80.0
Sella cart+Eckart	2.3	3.7	6.3	16.4	58.9	113.2
Sella cart no-Eckart	2.2	3.5	6.2	17.2	55.0	119.6
Sella internal	4.3	5.6	10.3	46.7	117.4	151.7

Valid TSs per GPU-hour (local gate)

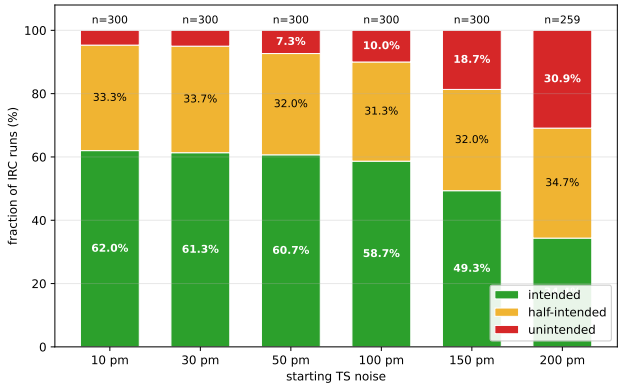
Method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD Eckart	227	158	115	70	40	22
GAD no-Eckart	168	115	84	56	30	16.5
Sella cart+Eckart	**264**	**249**	**175**	79	24	8.6
Sella cart no-Eckart	267	252	177	75	26	7.7
Sella internal	148	116	84	31	10	4

Crossover is at ∼100 pm. Below: Sella cartesian is fastest source of valid TSs. Above: GAD — the Sella convergence-rate collapse dominates per-hour yield.

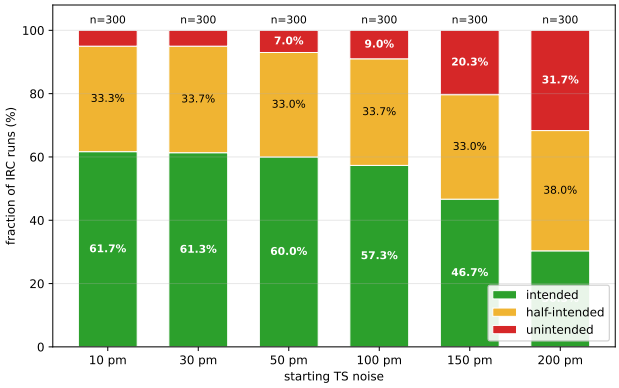
6 Appendix

6.1 Per-method bar charts: RMSD and endpoint vibrational quality

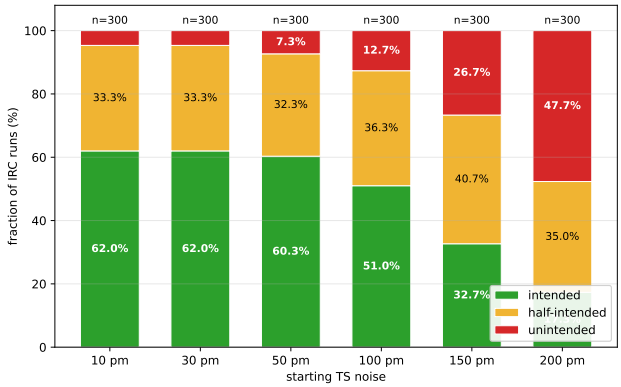
Three criteria per method: TOPO (already shown in §2.2), strict RMSD (< 0.3 Å), endpoint vibrational quality (both / one / neither at vibrational minimum).



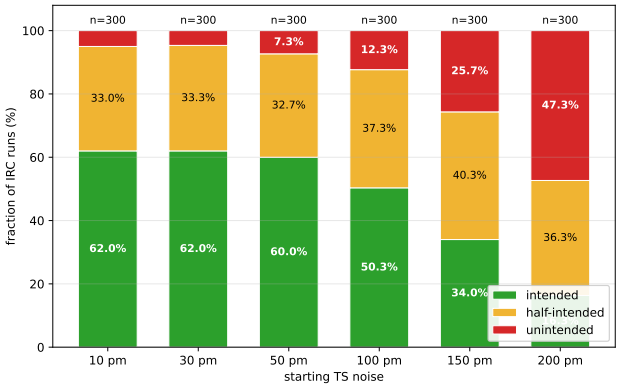
(a) GAD Eckart, RMSD



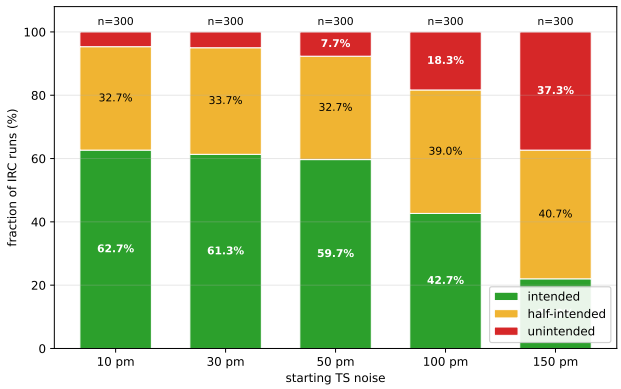
(b) GAD no-Eckart, RMSD



(c) Sella cart+Eckart, RMSD

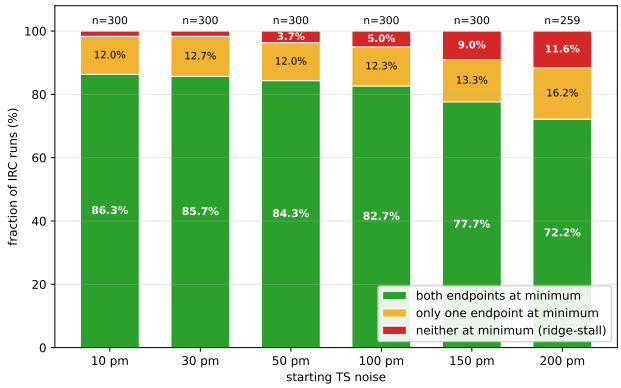


(d) Sella cart no-Eckart, RMSD

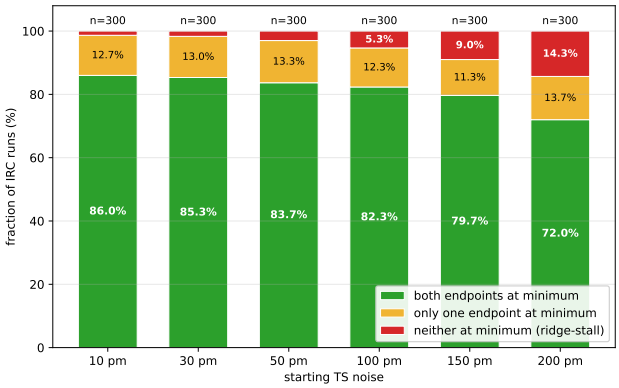


(e) Sella internal, RMSD

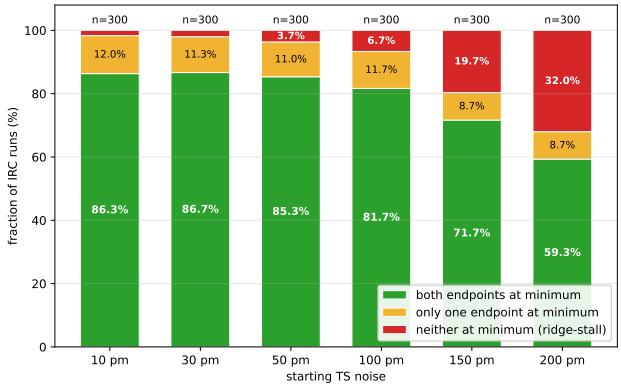
Figure 8: Strict RMSD criterion (< 0.3 Å) per method. Denominator = 300 samples per cell.



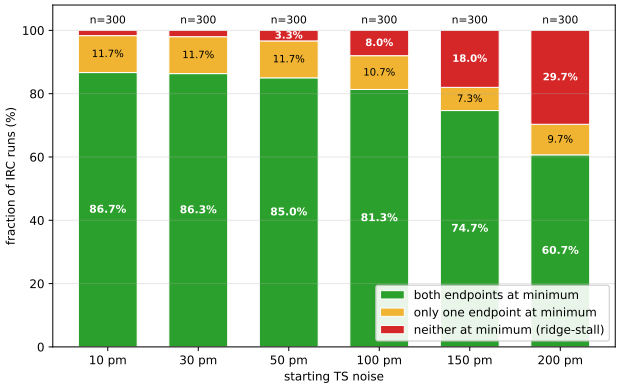
(a) GAD Eckart



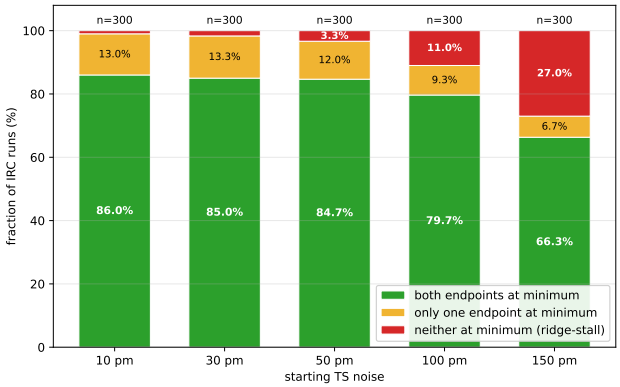
(b) GAD no-Eckart



(c) Sella cart+Eckart



(d) Sella cart no-Eckart



(e) Sella internal

Figure 9: Endpoint vibrational quality per method. Both / one / neither IRC endpoint reaches a true minimum ($n_{\mathrm{neg,vib}}$=0 on Eckart-projected Hessian).

6.2 All-three-classes line charts (cluttered, appendix only)

The main narrative uses "intended only" lines for readability. For completeness, the "intended + half + unintended" breakdown per method:

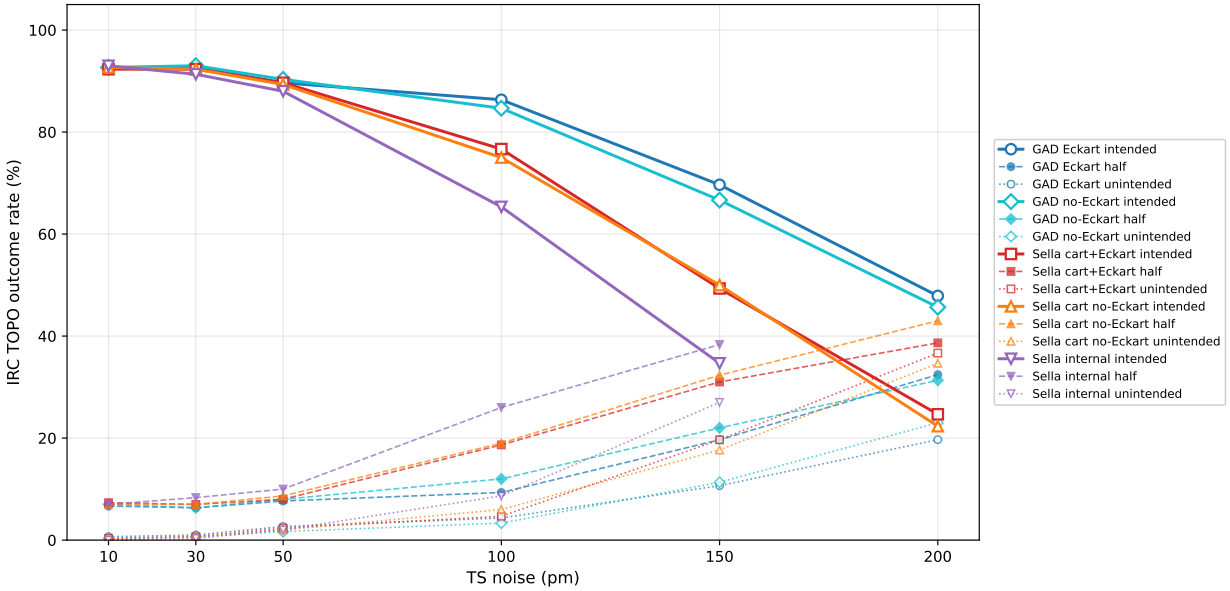


Figure 10: IRC TOPO outcomes — all three classes, all five methods. 15 lines total: solid = intended, dashed = half-intended, dotted = unintended. Cluttered on purpose; see §2.1 for the uncluttered headline view.

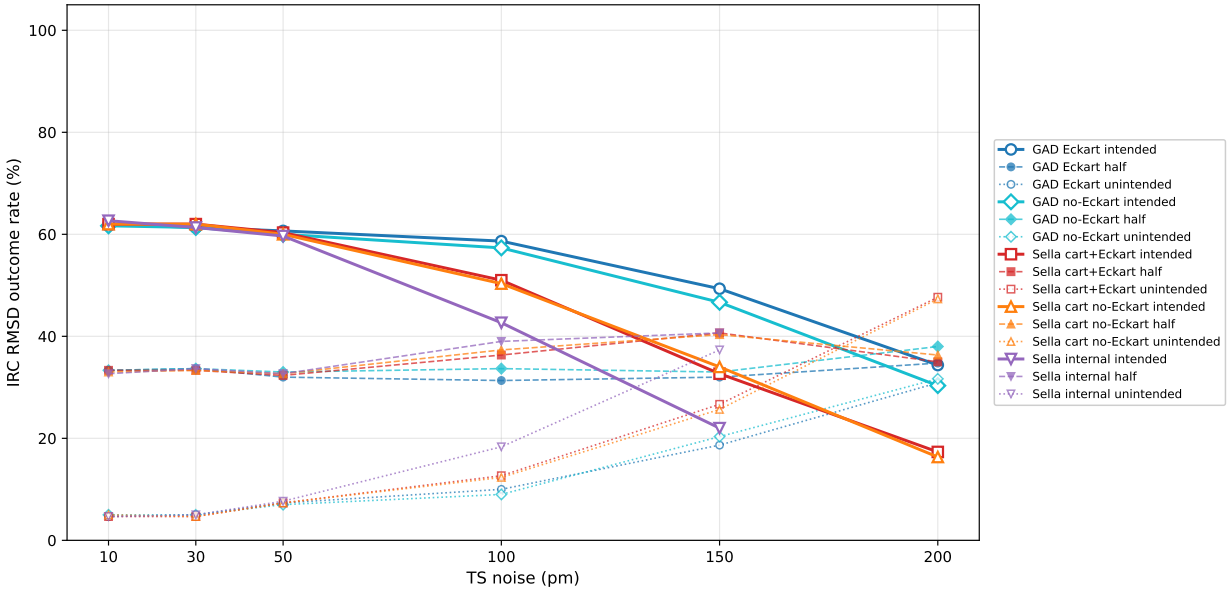


Figure 11: IRC strict-RMSD outcomes — all three classes, all five methods. Same layout, RMSD criterion (< 0.3 Å).

6.3 Sella IRC halt-at-step-0 fix

Initial Sella→IRC runs on Sella-refined TSs returned trivial results (IRC halted at step 0, wall ≈0.24 s per sample). Root cause: ASE's `Optimizer.irun` checks `gradient_converged` before any `step()`, and Sella's IRC halt threshold (fmax<0.01) matches Sella's TS optimizer threshold exactly. So a Sella-refined TS trivially satisfied the halt before the first kick.

Fix: `src/gadplus/search/irc_sella_hip.py::_force_first_kick` monkey-patches `converged`, `gradient_converg` and `step` so the first convergence check always returns False. Post-first-step behavior is unchanged.

6.4 Data locations

Artifact	Path
GAD dt003 (Eckart) summaries	`runs/round2`, `round3` (200 pm refilled)
GAD no-Eckart summaries	`runs/gad_no_eckart/`
Sella 2000-step (3 configs)	`runs/sella_2000/`
IRC on GAD Eckart	`runs/irc_sellahip_allendpoints/`
IRC on GAD no-Eckart	`runs/irc_gad_no_eckart/`
IRC on Sella cart+Eck 2000	`runs/irc_sella_carte_2000/`
IRC on Sella cart 2000	`runs/irc_sella_cart_2000/`
IRC on Sella internal 2000	`runs/irc_sella_int_2000/`
Figures	`figures/`
This report	`IRC_COMPREHENSIVE_2026-04-20.tex/pdf`

6.5 Open items

- Sella internal 200 pm IRC (final cell of the 5×6 matrix) still computing at report time.
- Rigorous predictor-corrector IRC sits at ∼52% TOPO regardless of noise (on GAD Eckart converged TSs). Backburner for investigation.
- Potential backfill of GAD Eckart numbers under $f_{\max}<0.01$ gate for strict apples-to-apples comparison (currently uses $\|\mathbf{F}\|_{\mathrm{mean}}$).